

Biofilm formation and virulence genes in Medical Devices-associated Bacterial Strains

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Introduction

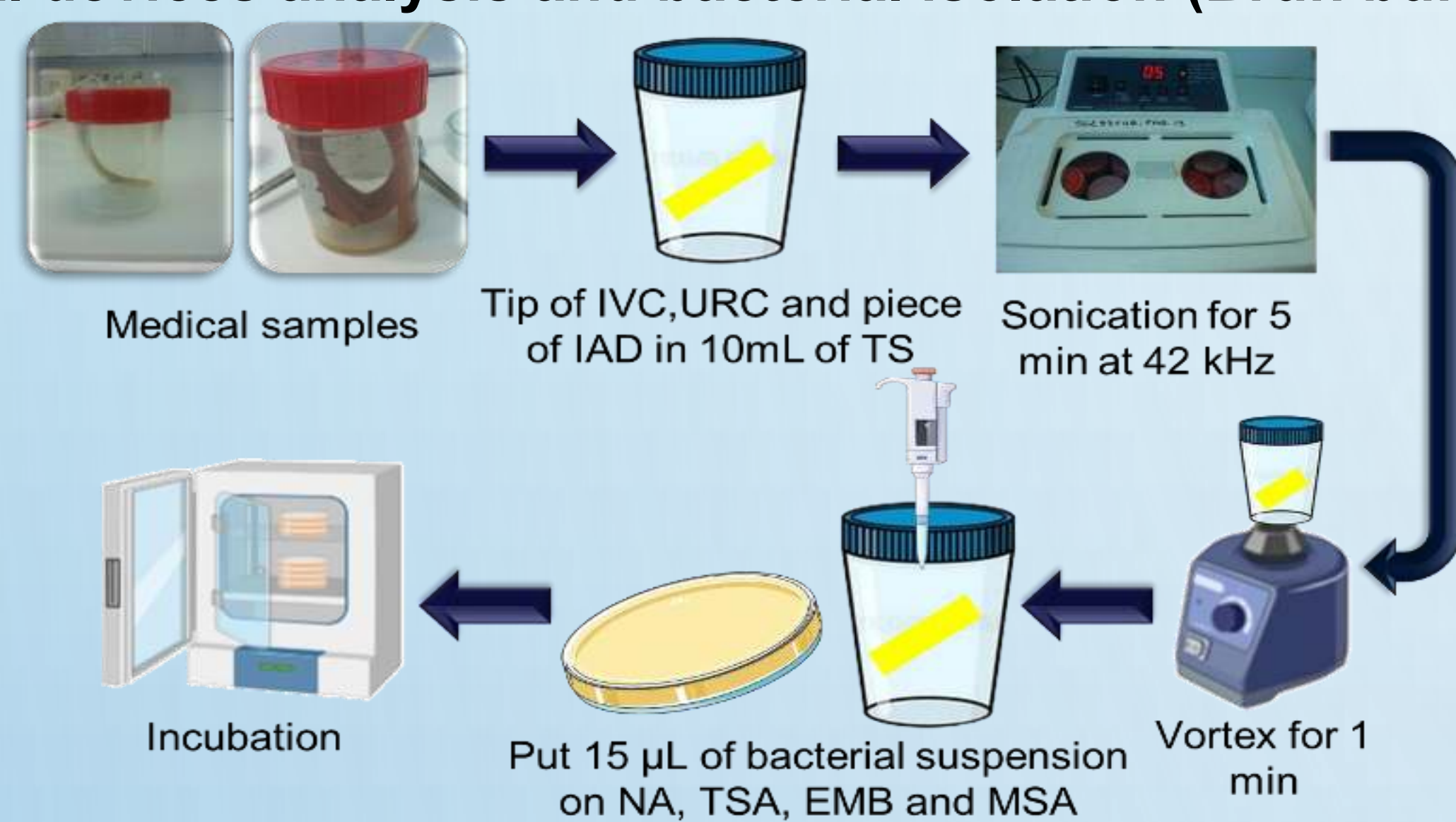
Every year, health providers implant millions of catheters to better treat acute and chronic illnesses in both adult and pediatric patients (Ullman *et al.*, 2015; Zhang *et al.*, 2020). Unfortunately, using them unavoidably makes it possible for the patient's own bacteria or accidental environmental contaminants to enter the underlying tissues and result in deadly complications (Casimero *et al.*, 2020, Aumeran *et al.*, 2021). The capacity of bacteria to attach to surfaces and form biofilms (Wi and Patel, 2018), as well as their persistence (Wang *et al.*, 2023), constitutes a real medical problem. As a result, antibiotic therapy is ineffective and biofilm eradication is impossible (Wainwright *et al.*, 2021).

Aim of the study

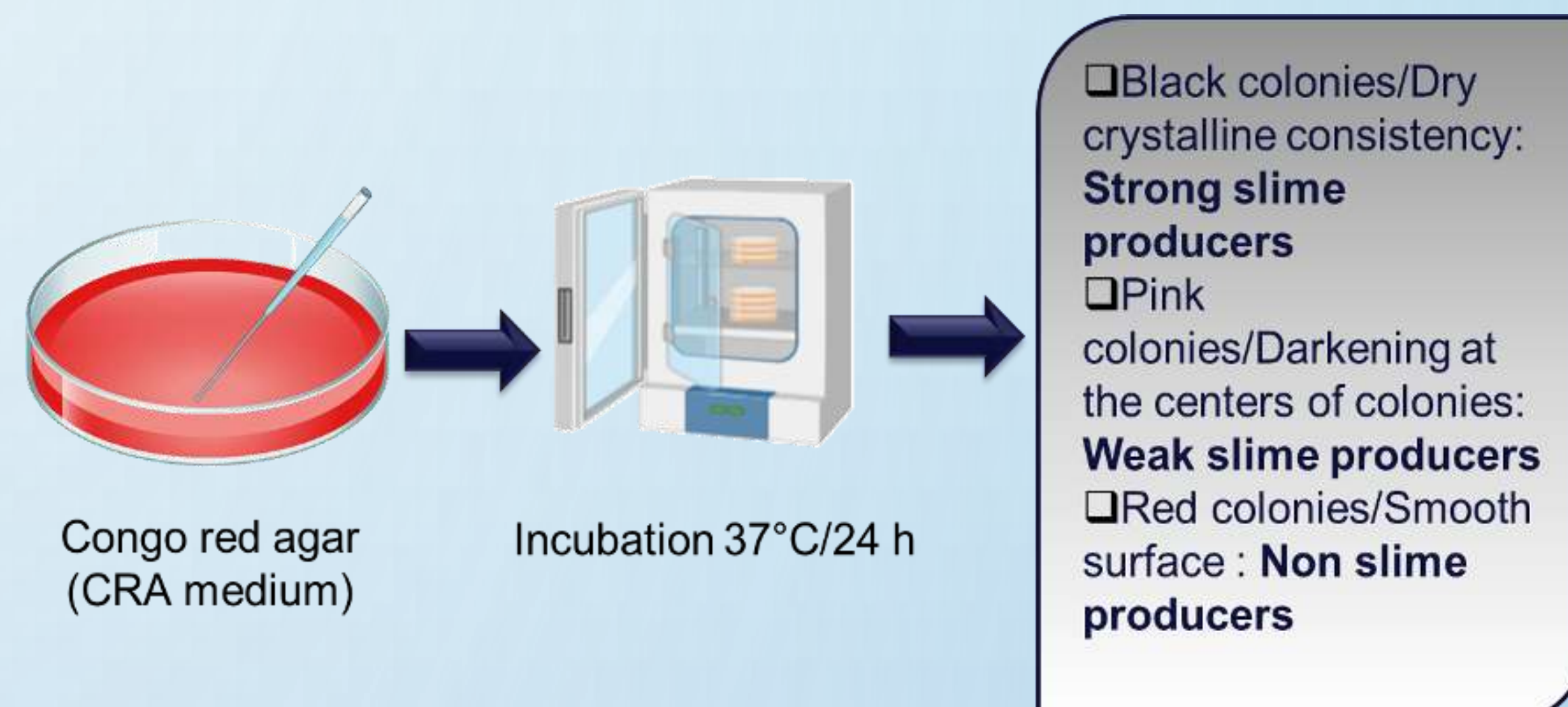
The study's goal is to characterize pathogenic bacteria that isolated from medical devices collected from patients hospitalized at least 48 h in different departments at two hospitals and a private healthcare unit (Bejaia, Algeria); in terms of their ability to form biofilms and pathogenicity (virulence gene content).

Materials and methods

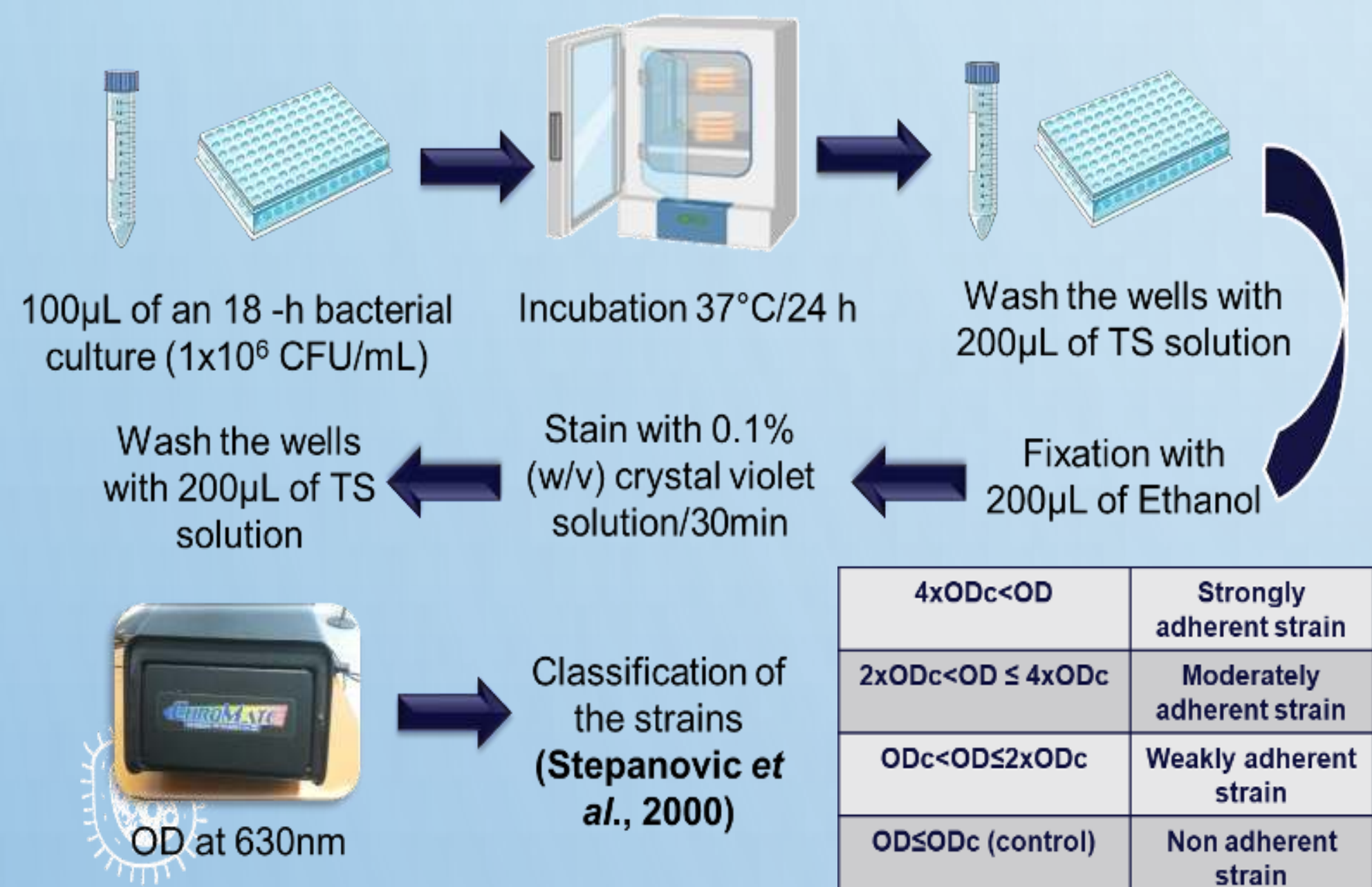
Medical devices analysis and bacterial isolation (Brun buisson *et al.* (1987)



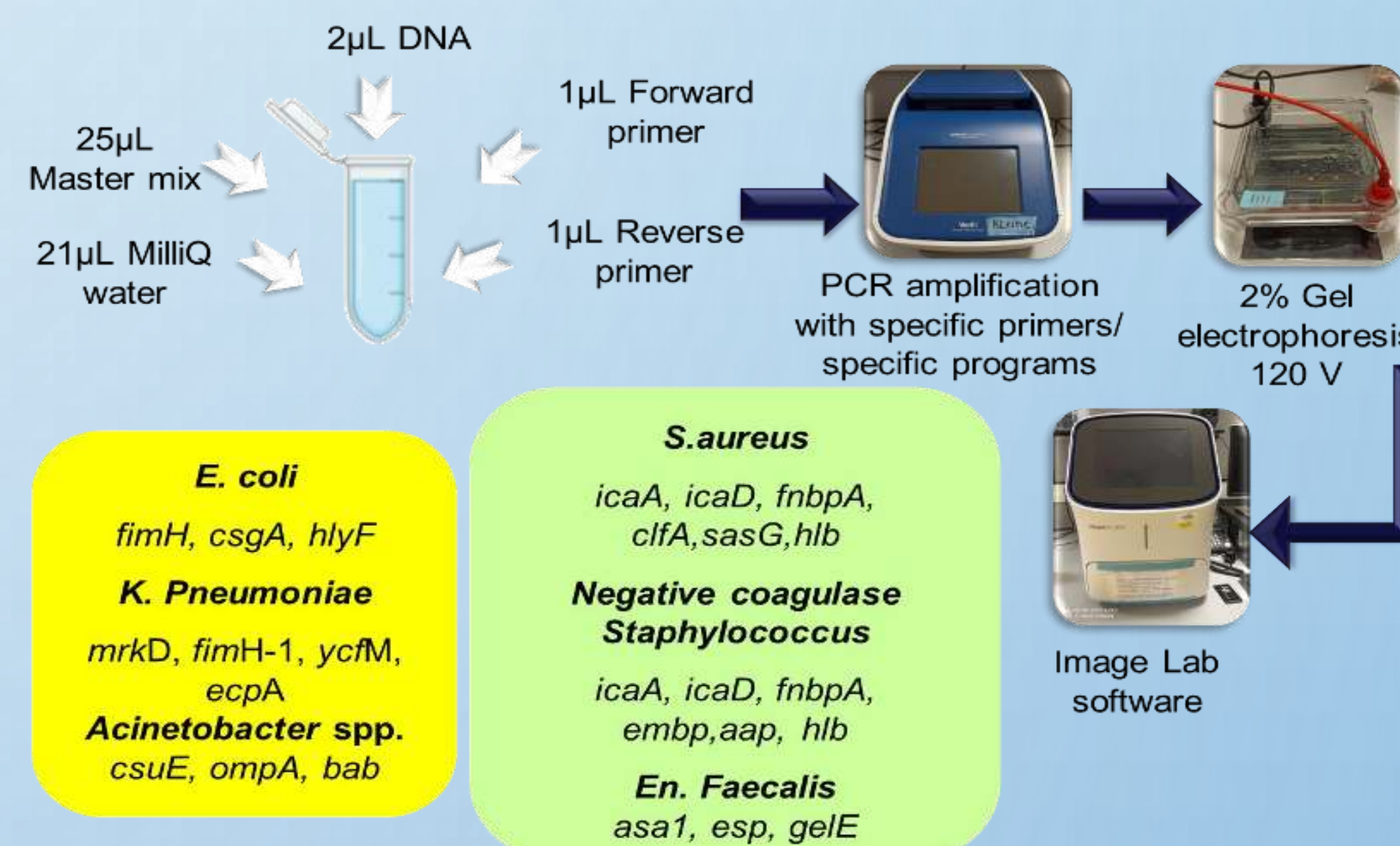
Slime production on Congo Red Agar (CRA) (Freeman *et al.*, 1989)



Biofilm formation on polystyrene plates (O'Tool *et al.*, 1998)



Detection of bacterial virulence and adhesion genes involved in biofilm formation



Results

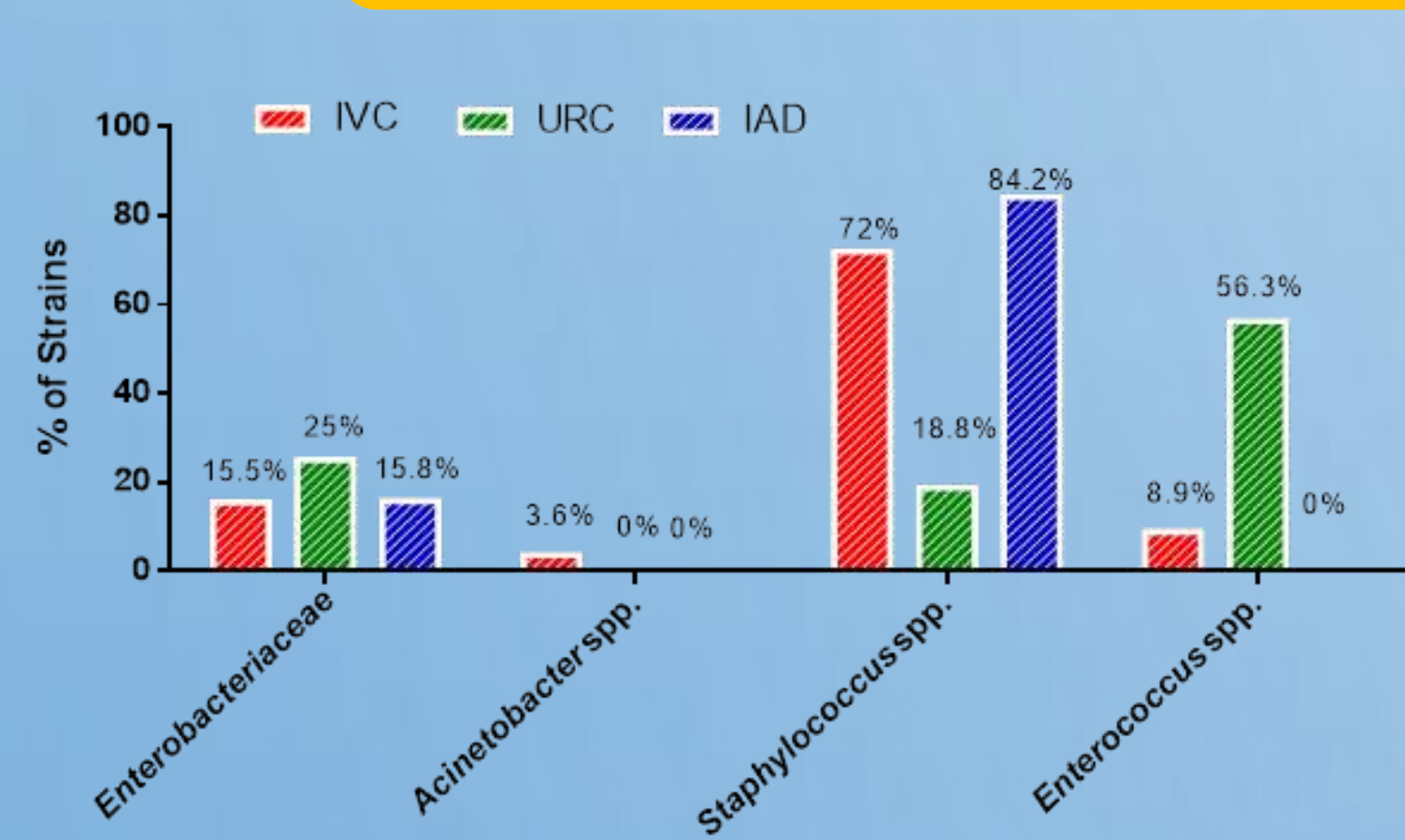


Figure 1. Distribution (%) of the isolates following the medical device.

Table 1. Proportion of biofilm-producing strains detected by TCP method.

	Biofilm formation (TCP assay)			
	OD > 0.12	0.06 < OD ≤ 0.12	0.03 < OD ≤ 0.06	OD ≤ 0.03
% of strains	Strong (%)	Moderate (%)	Weak (%)	Negative (%)
<i>E. coli</i>	85.4	5.94	5.02	3.65
<i>K. pneumoniae</i>	83.33	19.35	22.58	19.35
<i>Acinetobacter</i> spp.	33.33	16.67	33.33	16.67
<i>Staphylococcus</i> spp.	94.41	3.50	1.39	0.7
<i>En. faecalis</i>	100	0	0	0

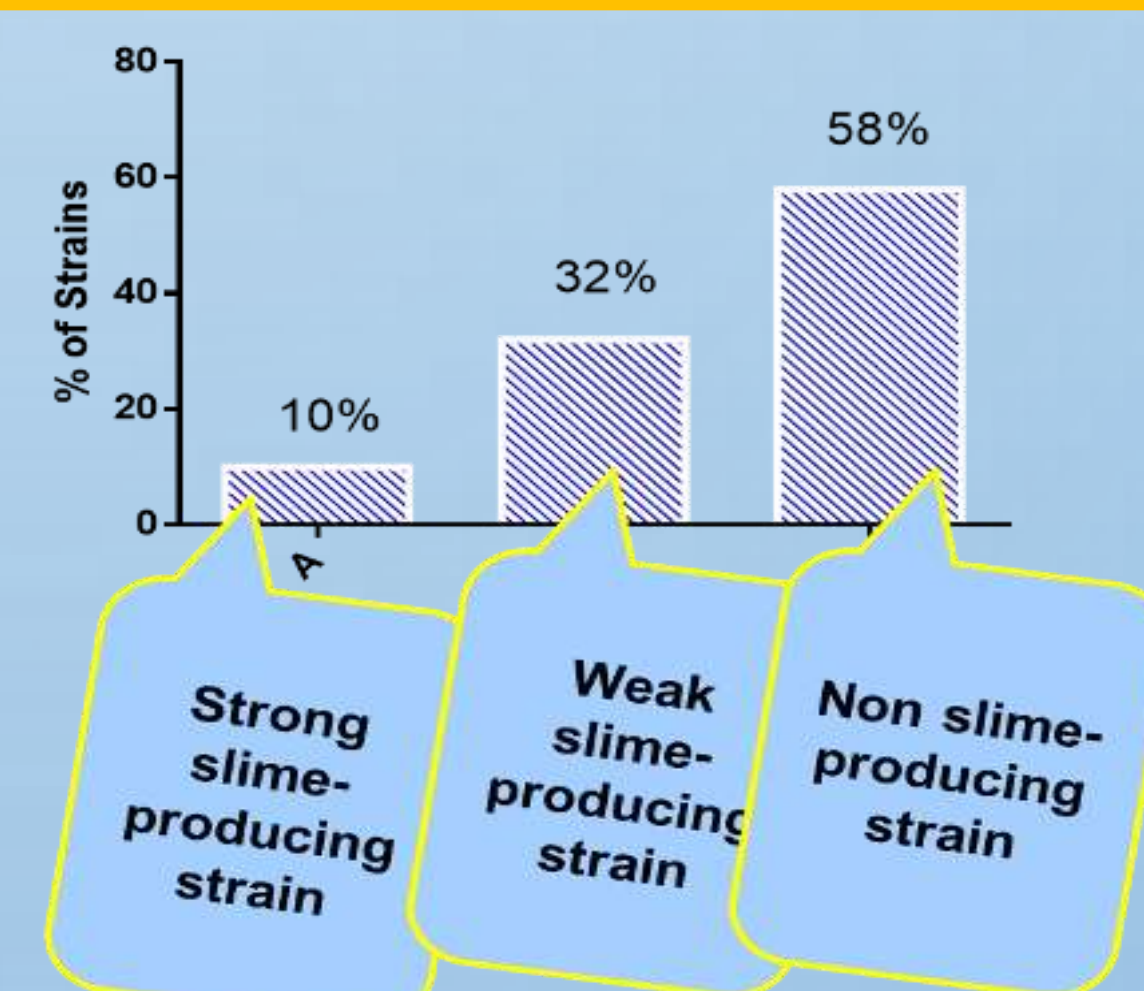


Figure 2. Proportion of slim-producing strains detected by CRA method.

Table 2. Rate of virulence genes in isolated strains.

Strain	Gene	Rate (%)	Strain	Gene	Rate (%)
<i>E. coli</i>	fimH	100	<i>Staphylococcus aureus</i>	icaA	83.3
	csgA	96.8		icaD	83.3
	hlyF	48.4		fnbA	100
<i>K. pneumoniae</i>	mrkD	100		clfA	100
	fimH	100		sasG	0
<i>Acinetobacter</i> spp.	ycfM	100	<i>Negative coagulase Staphylococcus</i>	hlb	100
	ecpA	100		icaA	34.8
	Bap	66.7		icaD	0
<i>En. faecalis</i>	csuE	83.3		fnbA	26.1
	ompA	100		embp	0
	esp	87.9		aap	0
	Asa1	93.9		hlb	21.7
	gelE	21.2			

Conclusion

- These medical devices are veritable ecological niches for various pathogenic bacteria that cause hospital-acquired infections.
- This current study demonstrated that the bacterial strains isolated from different medical devices were biofilm producers and harbored several virulence genes which contribute to bacterial persistence, pathogenicity, and treatment failure in medical device-related infections.

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